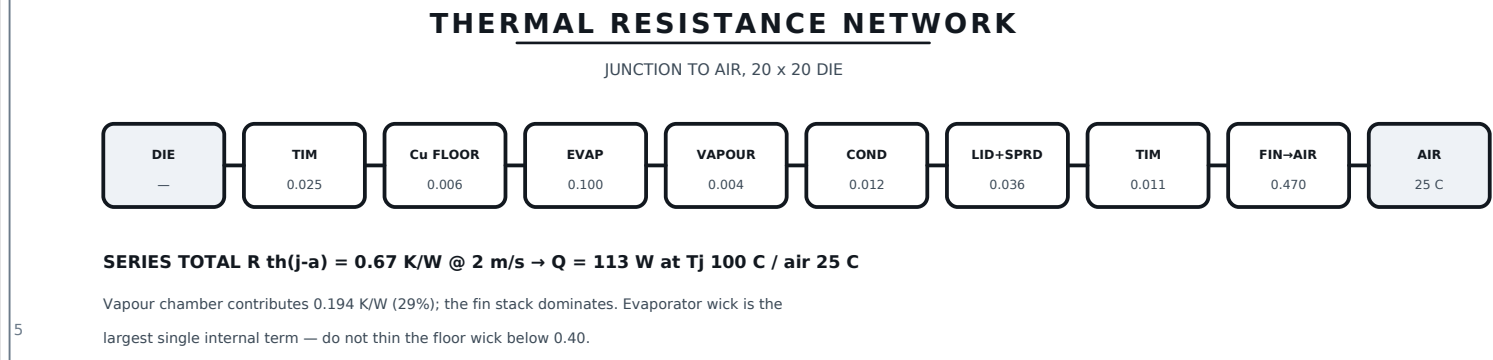
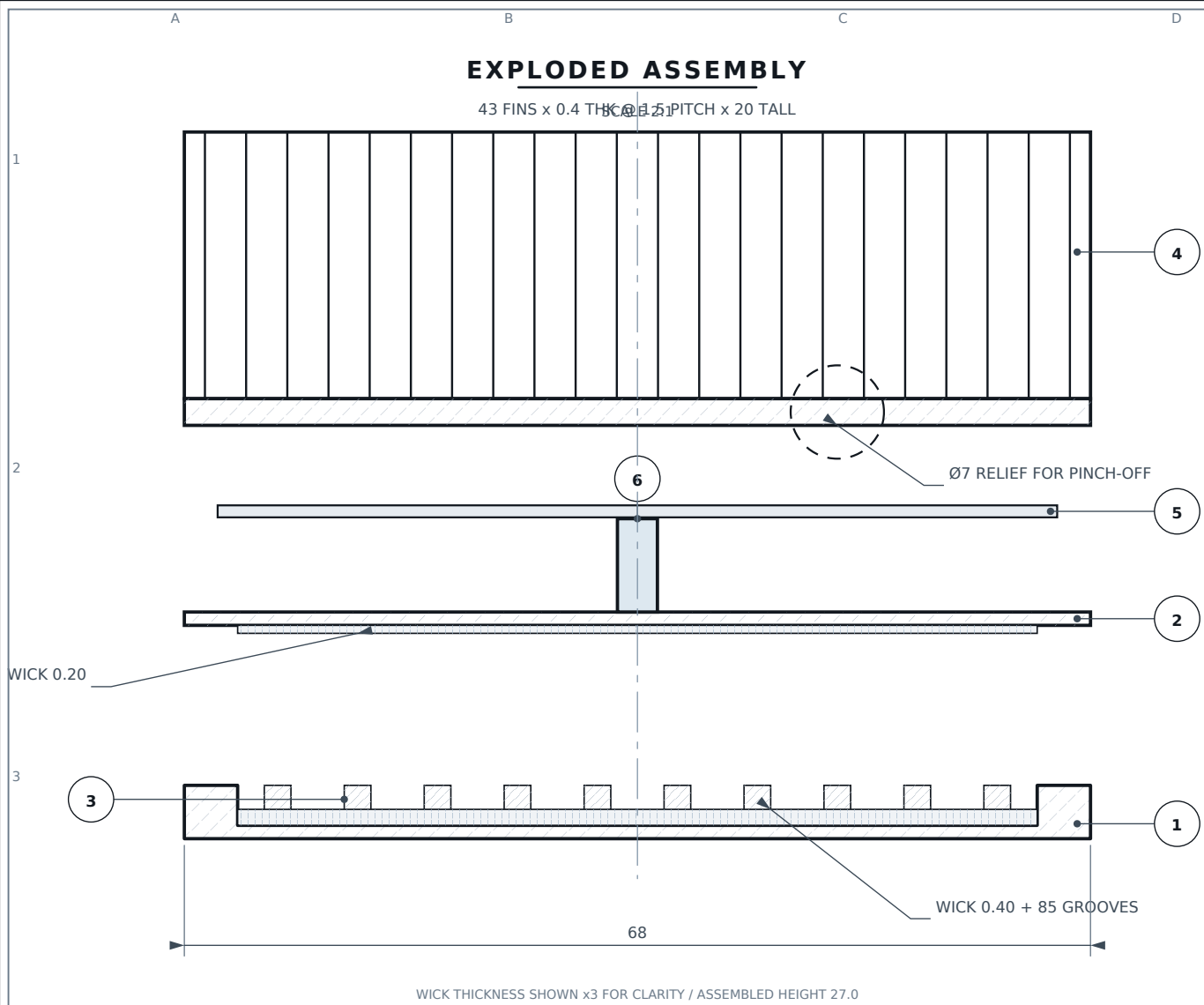


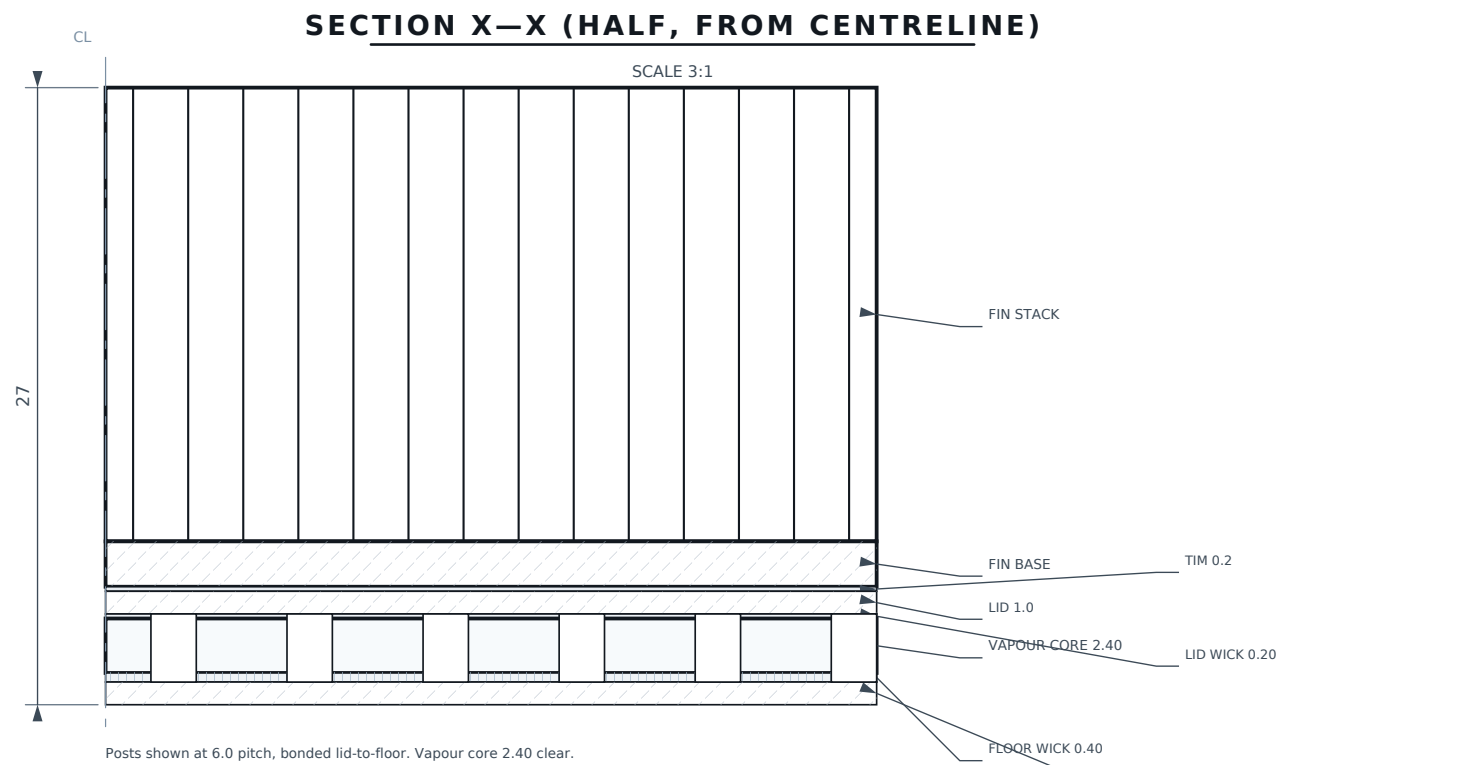


- GENERAL NOTES**
- Interpret per ISO 128. Tolerances ISO 2768-mK unless stated. Dimensions in mm.
 - Charging, evacuation and sealing per VC-300. NEVER charge before diffusion bonding — bond temperature exceeds the fluid critical pressure envelope.
 - CONFIG A requires forced air $\geq 1 \text{ m/s}$. For still air use CONFIG B (VC-202); a 1.5 mm fin pitch chokes natural convection to $h = 0.03 \text{ W/m}^2\text{K}$.
 - Orientation-independent: 10.5 kPa capillary head vs 0.66 kPa gravity head over 68 mm.

BILL OF MATERIALS — ASSEMBLY VC-100					
ITEM	QTY	PART No.	DESCRIPTION	MATERIAL	MASS
1	1	VC-101	CHAMBER BASE, POCKETED, 100 POSTS	Cu C10200 / C1020	77.4 g
2	1	VC-102	CHAMBER LID, 1.0 THK, FILL PORT	Cu C10200 / C1020	41.4 g
3	1	VC-103	SINTERED WICK — IN-SITU PROCESS	Cu POWDER 45-75um	9.6 g
4	1	VC-201	FIN STACK, 43 FIN — FORCED AIR	Al 1100-H14	88.1 g
4A	1	VC-202	FIN STACK, 13 FIN — NAT. CONVECTION	Al 1100-H14	72.7 g
5	1	VC-204	THERMAL INTERFACE PAD 0.2 THK	PCM / SIL-PAD	0.4 g
6	1	VC-205	PINCH-OFF TUBE OD3.0 x ID2.0 x 25	Cu C10200 / C1020	1.2 g
7	1.90 mL	VC-206	WORKING FLUID — DEAERATED	DI H2O 18 MOhm.cm	1.87 g

TOTAL MASS, CHARGED: 218 g (CONFIG A) / 203 g (CONFIG B)

PERFORMANCE — 20 x 20 DIE, $T_j 100 \text{ C}$, AIR IN 25 C			
CONFIGURATION	AIR	$R_{th(j-a)}$	$Q @ dT 75 \text{ K}$
CONFIG A — 43 fin @ 1.5 pitch	1 m/s	1.12 K/W	67 W
CONFIG A — 43 fin @ 1.5 pitch	2 m/s	0.67 K/W	113 W
CONFIG A — 43 fin @ 1.5 pitch	4 m/s	0.46 K/W	163 W
CONFIG B — 13 fin @ 5.0 pitch	natural	3.42 K/W	22 W
CONFIG B — passive, $T_j 85 \text{ C}$ limit	natural	—	16.5 W
capillary limit — water, vertical	—	—	804 W
capillary limit — water, horizontal	—	—	858 W



INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK	
TITLE GENERAL ASSEMBLY VAPOUR CHAMBER + FIN STACK / Cu-H2O / 68 x 68 x 27			SCALE 2:1		UNITS mm	
			THIRD ANGLE PROJECTION			
DRAWING No. VC-100			REV A	SHEET 1 / 7		
DRAWN	CHECKED	MATERIAL	FINISH		DATE	
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES		2026-09-09	